

Proving Properties of Parallelograms

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To prove properties of a parallelogram, always identify the appropriate **previously learned theorem**.

Opposite Sides Theorem

The **Opposite Sides Theorem** states that opposite sides of a parallelogram are congruent.

$$AB = CD \quad \text{and} \quad BC = AD$$

Opposite Angles Theorem

The **Opposite Angles Theorem** states that opposite angles of a parallelogram are congruent.

$$\angle A = \angle C \quad \text{and} \quad \angle B = \angle D$$

Consecutive Angles Theorem

The **Consecutive Angles Theorem** states that consecutive angles of a parallelogram are supplementary.

$$\angle A + \angle B = 180^\circ$$

Diagonals Bisect Each Other

The **Diagonals Bisect Each Other Theorem** states that the diagonals intersect at their common midpoint.

If diagonals AC and BD intersect at E , then

$$AE = EC \quad \text{and} \quad BE = ED$$

Diagonal Division Theorem

The **Diagonal Division Theorem** states that each diagonal divides a parallelogram into two congruent triangles.

For example, diagonal AC divides parallelogram $ABCD$ into

$$\triangle ABC \cong \triangle CDA$$

Examples

Example 1

In parallelogram $ABCD$, $AB = 18$ cm.

Therefore, $CD = 18$ cm.

The theorem used is the **Opposite Sides Theorem**.

Example 2

In parallelogram $EFGH$, $\angle E = 72^\circ$. Therefore, $\angle G = 72^\circ$.

The theorem used is the **Opposite Angles Theorem**.

Example 3

In parallelogram $JKLM$, $\angle J = 118^\circ$. Therefore, $\angle K = 62^\circ$.

The theorem used is the **Consecutive Angles Theorem**.

Example 4

Diagonals PR and QS of parallelogram $PQRS$ intersect at T . Therefore, $PT = TR$.
The theorem used is the **Diagonals Bisect Each Other Theorem**.

Example 5

In parallelogram $WXYZ$, diagonal WY is drawn. Therefore,

$$\triangle WXY \cong \triangle WZY$$

The theorem used is the **Diagonal Division Theorem**.

Practice Exercises

Practice 1

In parallelogram $ABCD$, $AB = 18$ cm.

Therefore, $CD = 18$ cm.

Identify the theorem that justifies the statement.

Practice 2

In parallelogram $EFGH$, $\angle E = 72^\circ$. Therefore, $\angle G = 72^\circ$.

Identify the theorem that justifies the statement.

Practice 3

In parallelogram $JKLM$, $\angle J = 118^\circ$. Therefore, $\angle K = 62^\circ$.

Identify the theorem that justifies the statement.

Practice 4

Diagonals PR and QS of parallelogram $PQRS$ intersect at T . Therefore, $PT = TR$. Identify the theorem that justifies the statement.

Practice 5

In parallelogram $WXYZ$, diagonal WY is drawn. Therefore,

$$\triangle WXY \cong \triangle WZY$$

Identify the theorem that justifies the statement.

Practice 6

In parallelogram $ABCD$, $BC = 25$ cm.

Therefore, $AD = 25$ cm.

Identify the theorem that justifies the statement.

Practice 7

In parallelogram $LMNO$, $\angle N = 105^\circ$.

Therefore, $\angle L = 105^\circ$.

Identify the theorem that justifies the statement.

Practice 8

In parallelogram $QRST$, $\angle R = 84^\circ$. Therefore, $\angle Q = 96^\circ$.

Identify the theorem that justifies the statement.

Practice 9

Diagonals AC and BD of parallelogram $ABCD$ intersect at E . Therefore, $BE = ED$.

Identify the theorem that justifies the statement.

Practice 10

In parallelogram $KLMN$, diagonal KM is drawn. Therefore,

$$\triangle KLM \cong \triangle KMN$$

Identify the theorem that justifies the statement.

Activities

Activity 1

In parallelogram $PQRS$, $PQ = 31$ cm.

Therefore, $RS = 31$ cm.

Justify the statement by identifying the theorem used.

Activity 2

In parallelogram $ABCD$, $\angle B = 54^\circ$. Therefore, $\angle D = 54^\circ$.

Justify the statement by identifying the theorem used.

Activity 3

In parallelogram $EFGH$, $\angle H = 127^\circ$. Therefore, $\angle E = 53^\circ$.

Justify the statement by identifying the theorem used.

Activity 4

Diagonals LN and MO of parallelogram $LMNO$ intersect at P . Therefore, $LP = PN$.

Justify the statement by identifying the theorem used.

Activity 5

In parallelogram $RSTU$, diagonal RT is drawn. Therefore,

$$\triangle RST \cong \triangle RUT$$

Justify the statement by identifying the theorem used.

Activity 6

In parallelogram $WXYZ$, $YZ = 16$ cm.

Therefore, $WX = 16$ cm.

Justify the statement by identifying the theorem used.

Activity 7

In parallelogram $JKLM$, $\angle K = 98^\circ$. Therefore, $\angle M = 98^\circ$.

Justify the statement by identifying the theorem used.

Activity 8

In parallelogram $CDEF$, $\angle C = 69^\circ$. Therefore, $\angle D = 111^\circ$.

Justify the statement by identifying the theorem used.

Activity 9

Diagonals VX and WY of parallelogram $VWXY$ intersect at Z . Therefore, $WZ = ZY$.

Justify the statement by identifying the theorem used.

Activity 10

In parallelogram $ABEF$, diagonal AE is drawn. Therefore,

$$\triangle ABE \cong \triangle AFE$$

Justify the statement by identifying the theorem used.

Thank you!